

PROFESSIONAL SUMMARY

Full Stack Python & AI Developer with 5+ years of experience building scalable backend systems and advanced AI solutions. Expert in Python, Flask, FastAPI, AWS, and GCP, with practical experience in database optimization, data pipelines, API design, RAG architectures, LLM integrations, Model Context Protocol (MCP), and Docker/Kubernetes deployments. Completed the University of Alberta Software Design and Architecture specialization covering object-oriented design, design patterns, software architecture, and service-oriented architecture.

PROFESSIONAL EXPERIENCE

Full Stack Developer & AI Engineer — **Freelance**

September 2020 – Present

- Built Vector RAG systems with Pinecone and hybrid search, and Graph RAG solutions with Neo4j and AWS Neptune.
- Developed Model Context Protocol integrations and dynamic routing between GPT-4, Claude, and LangChain workflows.
- Built prompt-chaining, LLM benchmarking, and audit-logging tools; deployed containerized services on AWS/GCP with Grafana and Prometheus.

Back End Developer — **Pivony**

December 2023 – August 2024

- Led an API service migration using Python and AWS DynamoDB and developed data pipelines with AWS EC2, AWS Glue, and Pandas.
- Optimized Flask/PostgreSQL endpoints and collaborated across backend and frontend teams on integration and deployment quality.

Full Stack Developer & Team Lead — **Linked Matrix International**

May 2020 – December 2023

- Built Vue.js, React, and Next.js interfaces; designed and optimized MySQL, PostgreSQL, MongoDB, and Snowflake data systems.
- Delivered AWS/GCP solutions and Kubernetes deployments with GitHub Actions and Helm. Named Best Employee of the Year in February 2023.

Associate Software Engineer — **Orange Lynx IT**

May 2019 – September 2020

- Delivered full-stack features for the Yakbun social application using Vue.js and PHP Laravel, with Kubernetes and GitHub Actions CI/CD.

TECHNICAL SKILLS

AI & LLM: Vector/Graph RAG, LangChain, Model Context Protocol, advanced prompt engineering, generative AI applications, OpenAI, Claude, GPT-4, Neo4j	Software Architecture: Service-oriented architecture, object-oriented design, design patterns, reusable software design, data structures, algorithms
Web: Python, FastAPI, Flask, Django, FastMCP, JavaScript, React/React 19, Next.js, Vue.js, WebSockets, SSE	Data: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Snowflake, Databricks, dbt, Pandas, AWS Glue
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Helm, ArgoCD, GitHub/GitLab CI, Grafana, Prometheus	Engineering: API design, system migrations, performance optimization, CI/CD, TDD, code review, Agile/Scrum

CERTIFICATIONS

Software Design and Architecture University of Alberta · Specialization · Aug 2, 2026 · 6BDEC1ZB5UIC	Service-Oriented Architecture University of Alberta · Aug 2, 2026 · R7QIAH5HZ9TI
Software Architecture University of Alberta · Jul 27, 2026 · 4EFPXFH6L50K	Design Patterns University of Alberta · Jul 27, 2026 · VXMWFKQKOP57

Programming with JavaScript

Meta · Jul 8, 2026 · [POPZ05Y01DLJ](#)

What's new in React 19?

Scrimba · Jan 14, 2026 · [Z2XS4IR6UU8B](#)

Object-Oriented Design

University of Alberta · Dec 3, 2025 · [7AJ58JVVYIO5](#)

Advanced Prompt Engineering for Everyone

Vanderbilt University · Oct 19, 2025 · [AO510SI5HAPV](#)

Data Structures and Algorithms: Merge Sort

Scrimba · Jun 29, 2026 · [5952NPRU3X7H](#)

Develop Generative AI Applications: Get Started

IBM · Dec 3, 2025 · [P0CME8E4FGUN](#)

Model Context Protocol (MCP) Mastery

Fractal Analytics · Oct 28, 2025 · [C8XMPTKB61QW](#)

EDUCATION

BSc, Computer Science

University of Engineering and Technology, Lahore

September 2016 – September 2020 · CGPA 3.48/4.00

RECOGNITION & VOLUNTEERING

Best Employee of the Year — Linked Matrix International, February 2023

United Nations Volunteers — Remote IT education support

Tea Community — Open-source contributor and mentor

Portfolio and certificate PDFs: asadrahim.github.io